STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Define stress. [1]
SI unit of Young’s modulus? [1]
Longitudinal strain? [1]
Hooke’s law? [1]
Modulus for volume change? [1]
Is strain dimensionless? [1]
Meaning of large Young’s modulus? [1]
Longitudinal stress? [1]

100 N on 2x10^-4 m^2: stress? [2]
2 m wire extends 1 mm: strain? [2]
Stress=2x10^7 Pa, strain=10^-4: Y? [2]
10 cm cube compresses by .1 cm: strain magnitude? [2]
L=1 m,A=2x10^-6 m^2,Y=2x10^11,F=40 N: extension? [2]
Why use steel reinforcement in concrete? [2]

Wire L=2m,A=10^-6,Y=2x10^11,F=100N: extension? [3]
Stress 4x10^7 Pa at strain 2x10^-4. Estimate modulus. [3]
B=1.5x10^9 Pa, volume strain=-2x10^-4. Pressure increase? [3]
Why does doubling wire length double its extension if F,A,Y fixed? [3]

Distinguish proportional and elastic limits. [4]
Derive extension and calculate for F=50 N,L=1.5m,A=10^-6,Y=2x10^11. [4]

End of paper - check working, units, signs and significant figures.